Name: Neel . P. Kankhare Class: SOC-8 Batch: A
Subject: IDS Roll No.: 23 Exp. No.:
Name of the Experiment:

Performed on:
Submitted on:

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10	

Q.1 Define an algorithm. list any four characteristics of an algorithm.

Ans:- An algorithm is a finite sequence of well defined, step-by-step instruction used to solve a specific problem or perform a computation

Characteristics :-

1. Input :- An algorithm takes zero or more inputs.
2. Output :- It produces at least one output
3. Definiteness :- Each step is clear, precise unambiguous.
4. Finiteness :- The algorithm terminals after a finite number of steps.

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Q.2) Define data structure and Abstract Data Type (ADT). Give one example of each.

Ans:-

A data structure is a way of organizing storing and managing data in memory as that if can be accessed and modified efficiently.

An Abstract Data Type (ADT) is a logical description of a data type Specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples:-

- Data Structure :- Array
- ADT :- Stack (operations :- push, pop, peek)

Q.3) Explain the process of converting a problem into a program Illustrate the role of algorithms in this process.

Ans:-

1. Problem Definition :- Understand requirement and Constraints

2. Analysis :- Identify inputs, outputs and processing

3. Algorithm Design :- Create step by step logic to solve the problem.

4. Flowchart / Pseudocode :- Represent the algorithm visually or textually.

5. Coding :- Convert algorithm into a program using a programming language.

6. Testing & Debugging :- Check Corrections and remove error.

7. Documentation & maintenance :- Improve and maintain the program.

Role of Algorithm :-
The algorithm acts as a blueprint of the solutions. It ensures correctness efficiency and makes coding easier and error - free.

Q.4 Explain linear and non-linear data structure with suitable examples.

Ans:-

Linear Data structure :-
Data elements are arranged sequentially
- Array
- Stack
- Queue
- linked list.

Non - linear Data structure :-
Data elements are arranged in a hierarchical or interconnected Manner.

Example :-
- Tree
- Graph.

Q.5 Differentiate b/w algorithm and program Analyze their role in SDLC.

Ans:-

Algorithm	Program
Step-by-step logic	Written Code.
language independent	language dependent
Design level	Implementation level

Algorithm	Program.
No Syntax rule	Follows Syntax rule.

Q.6) Examine the relationship among data, data Structure, and algorithm. justify how the choice of data structure affects algorithm efficiency.

Ans:-

Relationship :-

- Data is raw information
- Data Structure organizes the data.
- Algorithm processes the data using data structures.

Justification :-

The efficiency of an algorithm depends on the data Structure used.

- Searching is an array → slower
- Searching in a binary search here → faster (O (logn)).

Conclusion :-

Choosing the right data structure improves time and space efficiency of the algorithms.

Name: Neel . P. Kankhare Class: SOC-08 Batch: A
Subject: IDS Roll No.: 23 Exp. No.: 02
Name of the Experiment:

Performed on:
Submitted on:

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10	

1. Sequence organization and the characte-ristics of linear data structure.

Ans:- It is a method of storing data elements in a continuous memory location where elements are arranged one after another in a fixed order.

Characteristics :-
- Elements are stored in contiguous memory locations.
- Each elements has a unique index or position.
- Direct access to element is possible using indexing.
- Data is arranged in a linear sequence.
- Size is usually fixed.

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Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

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2. Array as an Abstract Data Type (ADT)

ANS:- ADT is collection of elements of the Same data type stored in contiguous memory location and accessed using an index.

- Components of ADT :-
  - Elements :- Data items of the Same type
  - Index :- Position used to access element
  - Size :- Total number of elements.

Basic operation on arrays :-
- Tranversal :- Visiting each elements
- Insesion :- Adding a new "
- Deletion :- Removing an "
- Searching :- Finding an "

3. Insesion, delection and Searching in a 1D array.

Ans:- Consider an array :-

A = [10, 20, 30, 40].

Insestion :- Insest 25 as position 2 :
Shift element right → A = [10, 20, 25, 30, 40]

Delection :- Delete elements at position 3
Shift elements left →
A = [10, 20, 25, 40]

Searching :- To Search 25
Compare Sequentially until found
25 is found at index 2.

4. Advantages and limitation of arrays.

Ans:- Advantages:-
- Past direct access using index (O(1))
- Simple and easy to implement
- Efficiency memory usage for fixed - size data.
- Good for storing homogeneous data.

limitations :- • Fixed size
• Insersion and deletation are time Consuming.
• Wastage of memory If size not used fully.

Efficiency Comparison :- Insertion :- O(n) due to shifting.
- Deletation :- O(n) due to shifting
- Searching :- O(n) for linear Search, O(log n) for binary search (Sorted array).

5. Two- dimensional array for students academic data:-

Ans:- 2D - array :- Rows represent Students
Coloums represent marks/Sub

Eg :-

Student	Math	Sci	English
S_1	85	90	88
S_2	78	82	80

Accessing data:- • Marks [0] [1] → Science Marks
• Tranversal uses nested loops

• Efficient Manipulation :-
• Direct indexing allows fast access
• Nested loops help in processing total and average.

• Eextension to n - dimensional array
• Image processing (3D array RGB)
• Scientics Stimulations.
• ML and data analytics.
• Games development.